

Engineering Mathematics: Cayley-Hamilton Theorem (3×3 Matrix)

This note provides a comprehensive guide to verifying the Cayley-Hamilton Theorem and finding the inverse of a 3×3 matrix.

1. Theoretical Foundation

The Cayley-Hamilton Theorem

The theorem states that every square matrix satisfies its own characteristic equation. For a 3×3 matrix A , if the characteristic equation is $\det(A - \lambda I) = -\lambda^3 + S_1\lambda^2 - S_2\lambda + S_3 = 0$, then:

$$-A^3 + S_1A^2 - S_2A + S_3I = 0$$

Conceptual Logic

- Characteristic Equation:** Determine the coefficients S_1 (Trace), S_2 (Sum of minors of diagonal elements), and S_3 (Determinant).
- Verification:** Substitute matrix A into the polynomial and show it results in a zero matrix.
- Inverse Computation:** Manipulate the equation to solve for A^{-1} using A^2 , A , and I .

2. Step-by-Step Verification & Application

Given the matrix A :

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

Step 1: Find the Characteristic Equation

We use the shortcut for 3×3 matrices: $\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$.

- S_1 (Trace): $1 + 2 + 3 = 6$
- S_2 (Sum of diagonal minors):

$$\begin{vmatrix} 2 & 1 \\ 0 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} = (6 - 0) + (3 - 4) + (2 - 0) = 6 - 1 + 2 = 7$$

- S_3 (Determinant $|A|$):
 $1(6 - 0) - 0 + 2(0 - 4) = 6 - 8 = -1$

Characteristic Equation:

$$\lambda^3 - 6\lambda^2 + 7\lambda + 1 = 0$$

Step 2: Verification of the Theorem

We must show that $A^3 - 6A^2 + 7A + I = 0$.

Calculate A^2 :

$$A^2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \longrightarrow A^2 = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

Calculate A^3 :

$$A^3 = A^2 \cdot A = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \longrightarrow A^3 = \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix}$$

Substitute into $A^3 - 6A^2 + 7A + I$:

$$\begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - \begin{bmatrix} 30 & 0 & 48 \\ 12 & 24 & 30 \\ 48 & 0 & 78 \end{bmatrix} + \begin{bmatrix} 7 & 0 & 14 \\ 0 & 14 & 7 \\ 14 & 0 & 21 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 3: Find the Inverse Matrix (A^{-1})

From $A^3 - 6A^2 + 7A + I = 0$, multiply by A^{-1} :

$$A^2 - 6A + 7I + A^{-1} = 0$$

$$A^{-1} = 6A - A^2 - 7I$$

Calculation:

$$A^{-1} = \begin{bmatrix} 6 & 0 & 12 \\ 0 & 12 & 6 \\ 12 & 0 & 18 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} - \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 7 \end{bmatrix}$$
$$A^{-1} = \begin{bmatrix} 6-5-7 & 0-0-0 & 12-8-0 \\ 0-2-0 & 12-4-7 & 6-5-0 \\ 12-8-0 & 0-0-0 & 18-13-7 \end{bmatrix} \longrightarrow A^{-1} = \begin{bmatrix} -6 & 0 & 4 \\ -2 & 1 & 1 \\ 4 & 0 & -2 \end{bmatrix}$$